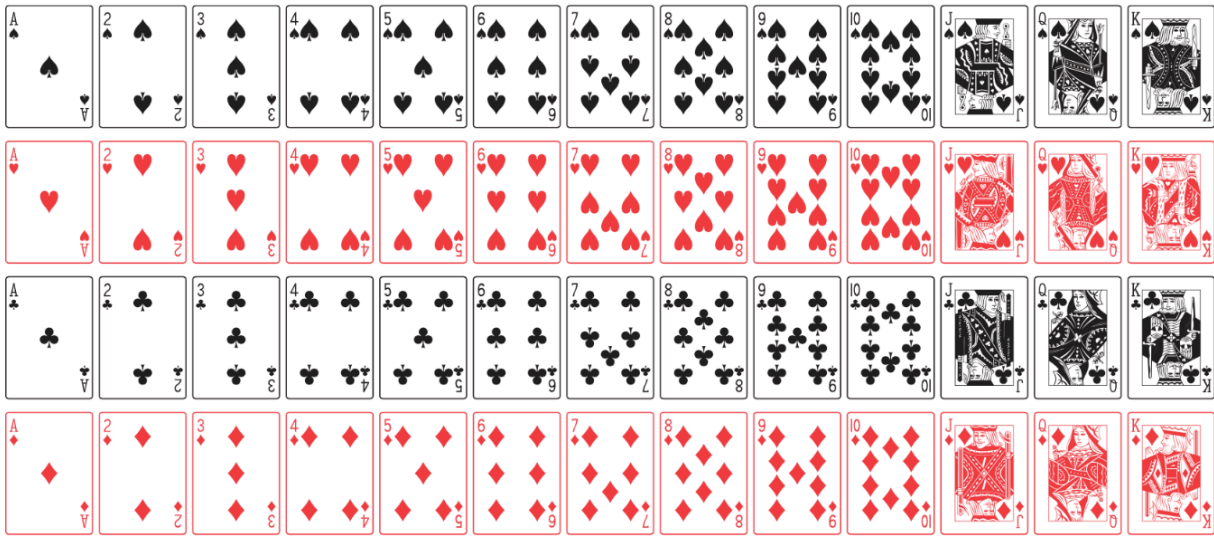


NAME:

STUDENT ID:

Consider a standard deck of 52 cards, as displayed in the following figure. In the first row (suit: spades ♠) and third row (suit: clubs ♣) all cards are black; in the second row (suit: hearts ♥) and fourth row (suit: diamonds ♦), all cards are red.



Consider the following events when a card is randomly selected.

- $A$ : card selected is a heart.
- $B$ : card selected is the color black (the spade ♠ and club ♣ cards are black).
- $C$ : card selected is a face card (Jack (J), Queen (Q), King (K)). Note that Ace (A) does **not** count as a face card.)
- $D$ : card selected is a King (K).

1. Find  $P(A)$ .

2. Find  $P(C)$ .

3. Find  $P(D^C)$ .

4. Find  $P(A \cup B)$ . Recall that if  $A$  and  $B$  have outcomes in common, you should only count them **once**.

5. Create a box model to represent drawing a 3 or an 8 in a single draw from the deck of cards described above.

Consider a *seven* sided die (with sides 0, 1, 2, 3, 4, 5 and 6). You roll this die once. Let:

- A be the event that a given roll yields an even number (0 is considered an even number).
  - B be the event that a given roll is greater than or equal to three.
  - C be the event that the number appears in the phrase “Stat 20”.
6. Find  $P(A)$ ,  $P(B)$  and  $P(C)$ . (**Hint:** write out the events *explicitly* and use this to compute their probabilities. That will help with all three questions.)

7. Find  $P(B \cap C)$

8. Find  $P(A \cap B^C)$ .

One number will be drawn at random from each of the two sets below:

$$A = \{1, 2, 3\}; B = \{1, 2, 3, 4\}$$

9. What is the probability the number drawn from  $A$  is greater than the one drawn from  $B$ ? (**Hint:** Write out the outcomes, for example  $(1, 1)$ ,  $(1, 2)$ , ....)

10. What is the probability that the number drawn from  $A$  is equal to the one drawn from  $B$ ?

Consider an outcome space  $\Omega$  with events  $A, B$  that are *not* mutually exclusive. Let  $P(A) = 0.5$ ,  $P(B) = 0.7$  and  $P(A \cap B) = .4$ . (Is it even *possible* for  $A$  and  $B$  to be mutually exclusive?)

11. Draw a Venn Diagram containing two circles, one representing  $A$  and the other  $B$ .

12. Shade in the space in the diagram above corresponding to event  $A$ .

13. Shade in the space in the diagram above corresponding to event  $B$  in a way that is different than the way you shaded in the space for event  $A$ .

14. Based on your shading, calculate the probability  $P(A \cup B)$ . Explain why you cannot add  $P(A)$  and  $P(B)$  directly to obtain this answer.